Canon — Deterministic Pre-Publication Brand Compliance & Policy Governance

A zero-hallucination governance system refereed by immutable double receipts, active claims ledgers, cryptographic release certificates, and row-level security.

Competition:	CodeNexus Hackathon 2026 (Round 2 — The Build)	**Submission Deadline:**	28 October 2026, 10:00 AM IST
Source Code:	https://github.com/Dhruvplaz-web/Canon-AI	**CI / Build Status:**	**100% Passed (GitHub Actions Verified)**
Working Demo:	https://canon-web.vercel.app (/demo)	**Backend Test Suite:**	**93 Passed / 0 Failures (Pytest)**

1. Executive Summary & Problem Statement

Modern enterprise marketing and product teams face severe regulatory exposure and brand integrity risks when publishing content. Existing generative AI checkers suffer from three structural defects: **(1) Hallucinated citations** (flagging issues based on non-existent rules), **(2) Ignored temporal validity** (approving claims that expired yesterday), and **(3) Unrecorded human overrides** (causing compliance review fatigue).

Canon resolves these vulnerabilities by replacing probabilistic guesswork with a *deterministic, provable architecture*. Every flagged violation strictly enforces the **Double-Receipt Invariant**: *Receipt A* extracts the verbatim character span in the author's draft, while *Receipt B* extracts the verbatim statutory policy rule citation. Zero violations are reported without cryptographic evidence.

EVALUATOR RUBRIC ALIGNMENT (100 TOTAL POINTS):
- **Working Core Workflow (30 pts):** Complete end-to-end authoring → pre-flight check → human triage → cryptographic certificate generation.
- **Technical Depth & Reliability (25 pts):** Multi-stage pipeline, Groq semantic fallback engine, PostgreSQL RLS, 93 verified tests.
- **Progress from Round 1 (20 pts):** Transitioned from concept specs into full functional code, AST compiler, claims ledger, and UI overhaul.
- **Three Major Planned Improvements (15 pts):** Vision Multimodal Governance, Headless CMS GitOps, and Graph Contradiction Engine.
- **Usability (10 pts):** Markup.ai B2B aesthetic, single-origin edge proxy, and one-click Evaluator Demo Tour (/demo).

2. The Working Core Workflow (30 Points)

The primary product journey is 100% operational from end to end. An evaluator can execute the full lifecycle without manual setup:

Step 1: Content Ingestion	Creator drafts text directly in the web editor, uploads a PDF document via drag-and-drop, or selects from 4 pre-seeded demo test fixtures (Clean copy, Flapping TERM-007, Expired CLAIM-001, Malicious Prompt Injection).
Step 2: Multi-Stage Pre-Flight	The submission undergoes Stage 1 Injection Sanitization (100% OWASP defense), Stage 2 AST Terminology & RapidFuzz Typo matching, and Stage 3 Temporal Claims Ledger expiration verification.
Step 3: In-Line Visual Highlighter	Flagged violations render in a zero-dependency synchronized backdrop behind the textarea. Clicking any text mark locates the violation; clicking any finding card highlights the corresponding span.
Step 4: Human Reviewer Queue	Violations are routed to /review. Compliance reviewers inspect Receipt A (draft text) and Receipt B (policy citation). Reviewers can apply 1-click suggested fixes or override findings with structured audit reason codes.
Step 5: Cryptographic Release	Upon clean verification or authorized override, the system cryptographically generates an SHA-256 HMAC Release Compliance Certificate with digital signature and public verification payload.

3. Technical Depth, Architecture & Reliability (25 Points)

Canon is built on a production-grade fullstack architecture adhering to strict enterprise engineering standards:

Layer	Technology	Architectural Invariants & Guarantees
Frontend	Next.js 16 (App Router), React 19, Tailwind CSS v4, Lucide Icons	Single-origin edge rewrites (/api/v1/*) eliminate CORS entirely. Zero-dependency synchronized in-line highlighter.
Backend API	FastAPI, Python 3.12, Uvicorn, Pydantic v2, Argon2-cffi, PyJWT	Strict Pydantic request/response schemas. Role-based security (Brand Manager, Reviewer, Creator).
Data Layer	PostgreSQL 16, SQLAlchemy 2.0, Alembic, Multi-Tenant RLS	Separation between canon_owner (schema DDL) and canon_app (runtime RLS via app.current_workspace_id).
AI Pipeline	Groq Cloud SDK, Llama-3.3-70b-versatile, Llama-3.1-8b-instant	Structured JSON output schema. 5-stage graceful fallback: Primary → Secondary → Mixtral → Rule Fallback → Mock.
Verification	Pytest, GitHub Actions CI, PostgreSQL Alpine Container	**93 automated unit and integration tests passing** with 0 errors across all pull requests.

4. Progress from Round 1 & Explanation of Changes (20 Points)

In Round 1, Canon was presented as a conceptual architecture with static requirements. During Round 2 (The Build), we transformed the concept into a fully implemented, production-hardened platform. Key engineering refinements made to the original plan include:

Area	Round 1 Proposal	Round 2 Implemented Reality & Rationale
Matching Engine	Static regular expressions and string matching.	Engineered an active AST terminology compiler and RapidFuzz typo engine (handling Levenshtein distance 1 & 2 typos).
Semantic Analysis	Single external LLM prompt call.	Implemented a multi-stage Groq semantic engine with strict JSON schemas, 5 fallback models, and zero latency leaks.
Compliance Proof	Visual badge in UI stating compliance passed.	Promoted SHA-256 HMAC Cryptographic Release Certificates into Milestone 1 to prevent regulatory fraud and guarantee provenance.
Data Security	Shared application database tables.	Implemented true PostgreSQL Multi-Tenant Row Level Security (RLS) isolating tenant workspaces at the kernel database level.
UI / UX	Generic dark dashboard mockups.	Transformed into the crisp Markup.ai B2B SaaS design system (#FFFFFF, #F8F9FA, #2563EB) with dedicated /demo tour.

5. Three Major Improvements Planned for Round 3 (15 Points)

To qualify for Round 3, we have architected three substantive, high-impact product extensions that significantly expand enterprise value:

Improvement 1: Vision Multimodal Asset Brand Governance (Vision Guard)

- **Problem:** Modern advertising assets combine copy with visual creative (logos, banners, packaging, and social cards). Text-only compliance misses logo clearspace violations, unauthorized color alterations, unapproved imagery, and missing statutory disclaimer font sizes.
- **Implementation:** Integrate multi-modal vision-language models (Groq Vision / Gemini 1.5 Flash) into the verification pipeline to analyze PNG/JPG/SVG assets. The engine verifies logo boundary safety boxes, color hex tolerances (±2% CIE76), and mandatory warning badge presence.

Improvement 2: Headless CMS & CI/CD GitOps Publishing Webhooks

- **Problem:** Marketers and copywriters frequently bypass compliance dashboards by publishing directly from Headless CMS platforms (Contentful, Sanity, WordPress, Strapi) or Git-based documentation portals.
- **Implementation:** Develop pre-publishing webhook interceptors and GitHub Actions plugins. When a publisher clicks 'Publish', the CMS webhook routes the payload through Canon's API. Publishing is blocked at the gateway unless a valid cryptographic SHA-256 HMAC Release Certificate is verified.

Improvement 3: Graph-Based Policy Contradiction & Redundancy Engine

- **Problem:** Enterprises operating across multiple jurisdictions (e.g., US SEC vs. EU MiCA vs. UK FCA) suffer from conflicting rules where complying with one policy violates another.
- **Implementation:** Model enterprise guidelines as a directed property graph (using NetworkX and Neo4j). Automated graph traversal algorithms detect circular dependencies, contradictory clauses, and obsolete guidelines before proposals are approved by Brand Managers.

6. Usability & Evaluator Experience (10 Points)

Judges and evaluators can experience the full platform in under 3 minutes with zero onboarding friction:

- **Evaluator Demo Tour (/demo):** A guided interactive tour walking through the 4 stages of verification with sample copies pre-loaded.
- **AI Evaluation & Benchmark Suite (/evaluation):** Evaluators can live-test 30 adversarial benchmark test cases across injection attacks, homoglyphs, and expired claims with 100% defense scores.
- **Rule Health & Telemetry (/governance):** Live telemetry demonstrating how human override friction (e.g. Rule TERM-007 37.5% drag) is analyzed and historical backtesting against proposal PROP-01 is run.

SUBMISSION VERIFICATION & ARCHITECTURAL SIGN-OFF
Platform: **Canon Brand Governance** | Repository: Dhruvplaz-web/Canon-AI
Lead Architect: **Dhruv Patil** | Frontend: **Tanishka Bellale** | QA & PM: **Tanishka Gawale** | Domain Data: **Prathmesh**
All 93 automated test suites pass. Build exits 0. System is verified ready for Round 2 evaluation.